

Force and Motion

MCQ

1. Force and motion are related because

- a) Motion exists without force
- b) Force can produce motion
- c) Motion stops force
- d) Force depends on mass only

Answer: b) Force can produce motion

2. Force is defined as

- a) Energy stored in an object
- b) Push or pull that changes velocity
- c) Mass times acceleration
- d) Ability to do work

Answer: b) Push or pull that changes velocity

3. Force can change all EXCEPT

- a) Shape
- b) Size
- c) Colour
- d) Direction

Answer: c) Colour

4. Force is a

- a) Scalar quantity
- b) Vector quantity
- c) Dimensionless quantity
- d) Constant quantity

Answer: b) Vector quantity

5. SI unit of force is

- a) Joule
- b) Watt
- c) Newton
- d) Pascal

Answer: c) Newton

6. Which is an example of force?

- a) Burning candle
- b) Man pushing a car
- c) Melting ice
- d) Rusting iron

Answer: b) Man pushing a car

7. Motion means

- a) Object at rest
- b) Change in position with time
- c) Increase in mass
- d) Change in colour

Answer: b) Change in position with time

8. Motion depends on

- a) Weight
- b) Shape
- c) Surroundings
- d) Temperature

Answer: c) Surroundings

9. Distance is

- a) Vector quantity

b) Scalar quantity

c) Always zero

d) Directional

Answer: b) Scalar quantity

10.Distance measures

a) Direction only

b) Shortest path

c) Length without direction

d) Change in velocity

Answer: c) Length without direction

11.Displacement is

a) Scalar

b) Directionless

c) Vector

d) Always positive

Answer: c) Vector

12.Displacement depends on

a) Total path

b) Direction and magnitude

c) Speed only

d) Time only

Answer: b) Direction and magnitude

13.Displacement is zero when

a) Distance is zero

b) Speed is zero

c) Initial and final positions are same

d) Time is zero

Answer: c) Initial and final positions are same

14.Speed is defined as

a) Distance $\times$ Time

b) Distance / Time

c) Time / Distance

d) Displacement / Time

Answer: b) Distance / Time

15.Speed tells us

a) Direction

b) Force

c) How fast an object moves

d) Acceleration

Answer: c) How fast an object moves

16.Speed is inversely proportional to

a) Distance

b) Force

c) Time

d) Acceleration

Answer: c) Time

17.Uniform speed means

a) Unequal distance in equal time

b) Equal distance in equal time

c) No motion

d) Changing direction

Answer: b) Equal distance in equal time

18. Non-uniform speed means

- a) Equal distance in equal time
- b) Equal distance in unequal time
- c) Unequal distance in equal time
- d) Constant motion

Answer: c) Unequal distance in equal time

19. Average speed is

- a) Maximum speed
- b) Minimum speed
- c) Total distance / Total time
- d) Displacement / Time

Answer: c) Total distance / Total time

20. Velocity differs from speed because it has

- a) Time
- b) Distance
- c) Direction
- d) Mass

Answer: c) Direction

21. Velocity is a

- a) Scalar
- b) Vector
- c) Constant
- d) Dimensionless

Answer: b) Vector

22. Uniform velocity means

- a) Equal speed only

- b) Equal displacement in equal time in same direction
- c) Changing direction
- d) Zero motion

Answer: b) Equal displacement in equal time in same direction

23. Light travelling in vacuum is an example of

- a) Non-uniform velocity
- b) Uniform velocity
- c) Non-uniform speed
- d) Rest

Answer: b) Uniform velocity

24. Non-uniform velocity occurs when

- a) Speed is constant
- b) Direction is constant
- c) Speed or direction changes
- d) Time is zero

Answer: c) Speed or direction changes

25. A train starting from station shows

- a) Uniform velocity
- b) Rest
- c) Non-uniform velocity
- d) Zero acceleration

Answer: c) Non-uniform velocity

26. Average velocity equals

- a) Distance / Time
- b) Speed $\times$ Time
- c) Displacement / Time

d) Distance $\times$ Speed

Answer: c) Displacement / Time

27. Acceleration is

a) Change in distance

b) Rate of change of velocity

c) Rate of change of distance

d) Speed / Distance

Answer: b) Rate of change of velocity

28. Acceleration is measured in

a) m/s

b) m/s²

c) km/h

d) Newton

Answer: b) m/s²

29. Acceleration is a

a) Scalar

b) Vector

c) Constant

d) Dimensionless

Answer: b) Vector

30. Positive acceleration means

a) Speed decreases

b) Velocity increases

c) Direction changes

d) Motion stops

Answer: b) Velocity increases

31. A van increasing its speed has

- a) Negative acceleration
- b) Zero acceleration
- c) Positive acceleration
- d) No motion

Answer: c) Positive acceleration

32. Negative acceleration is also called

- a) Speed
- b) Motion
- c) Deceleration
- d) Force

Answer: c) Deceleration

33. Applying brakes to a bike causes

- a) Positive acceleration
- b) Uniform acceleration
- c) Deceleration
- d) No acceleration

Answer: c) Deceleration

34. Uniform acceleration means

- a) Same velocity always
- b) Velocity changes equally in equal time
- c) Unequal velocity change
- d) Zero velocity

Answer: b) Velocity changes equally in equal time

35. Freely falling body shows

- a) Non-uniform acceleration

b) Uniform acceleration

c) No acceleration

d) Rest

Answer: b) Uniform acceleration

36. Raindrops falling show

a) Non-uniform motion

b) Uniform acceleration

c) Zero velocity

d) Neutral motion

Answer: b) Uniform acceleration

37. Non-uniform acceleration means

a) Same change in velocity

b) Unequal change in velocity

c) Zero acceleration

d) Constant speed

Answer: b) Unequal change in velocity

38. Circular motion shows

a) Uniform velocity

b) Non-uniform acceleration

c) Zero speed

d) Rest

Answer: b) Non-uniform acceleration

39. Distance–time graph shows

a) Force

b) Position of object

c) Mass

d) Weight

Answer: b) Position of object

40. In distance–time graph, time is on

a) y-axis

b) both axes

c) x-axis

d) origin

Answer: c) x-axis

41. Distance is plotted on

a) x-axis

b) y-axis

c) origin

d) scale

Answer: b) y-axis

42. Speed–time graph shows

a) Distance variation

b) Speed variation

c) Displacement

d) Acceleration only

Answer: b) Speed variation

43. In speed–time graph, steeper line means

a) Less speed

b) No speed

c) Greater speed

d) Zero acceleration

Answer: c) Greater speed

44.Distance–time graph helps to study

- a) Force
- b) Stability
- c) Uniform motion
- d) Temperature

Answer: c) Uniform motion

45.Speed–time graph is useful to

- a) Find mass
- b) Compare speeds
- c) Find shape
- d) Find colour

Answer: b) Compare speeds

46.Centre of mass is

- a) Highest point
- b) Lowest point
- c) Point where mass seems concentrated
- d) Edge of object

Answer: c) Point where mass seems concentrated

47.Centre of gravity is the point

- a) Where mass is zero
- b) Through which weight acts
- c) With no force
- d) At top of object

Answer: b) Through which weight acts

48.For regular objects, centre of gravity lies

- a) At edge

- b) Outside object
- c) At geometric centre
- d) At base

Answer: c) At geometric centre

49. An object balances when supported at

- a) Any point
- b) Centre of gravity
- c) Edge
- d) Base

Answer: b) Centre of gravity

50. Plumb line is used to find

- a) Mass
- b) Weight
- c) Centre of gravity
- d) Force

Answer: c) Centre of gravity

51. Centre of gravity of irregular body is

- a) Multiple points
- b) Only one point
- c) Outside body always
- d) Variable

Answer: b) Only one point

52. Stability means

- a) Ability to fall
- b) Ability to move
- c) Ability to maintain position

d) Ability to rotate

Answer: c) Ability to maintain position

53. Stable equilibrium occurs when

a) Object falls easily

b) C.G. lies outside base

c) C.G. lies within base

d) Base is small

Answer: c) C.G. lies within base

54. Large base gives

a) Less stability

b) No stability

c) More stability

d) Neutral stability

Answer: c) More stability

55. Unstable equilibrium occurs when

a) C.G. is low

b) C.G. is within base

c) C.G. is outside base

d) Base is large

Answer: c) C.G. is outside base

56. Small base results in

a) High stability

b) More balance

c) Less stability

d) No gravity

Answer: c) Less stability

57. Neutral equilibrium means

- a) Object returns to original position
- b) Object falls
- c) Object stays in new position
- d) Object stops moving

Answer: c) Object stays in new position

58. Lowering centre of gravity

- a) Decreases stability
- b) Increases stability
- c) No effect
- d) Removes gravity

Answer: b) Increases stability

59. Increasing base area

- a) Reduces stability
- b) Increases stability
- c) Stops motion
- d) Changes mass

Answer: b) Increases stability

60. Heavy base causes

- a) Higher C.G.
- b) Lower C.G.
- c) No gravity
- d) Unstable object

Answer: b) Lower C.G.

61. Thanjavur doll moves due to

- a) High speed

- b) High force
- c) Low centre of gravity
- d) Large height

Answer: c) Low centre of gravity

62. Thanjavur doll is made of

- a) Plastic
- b) Wood
- c) Terracotta
- d) Metal

Answer: c) Terracotta

63. Thanjavur doll movement is

- a) Linear
- b) Oscillatory
- c) Circular
- d) Random

Answer: b) Oscillatory

64. Velocity depends on

- a) Distance only
- b) Time only
- c) Direction and speed
- d) Mass

Answer: c) Direction and speed

65. Distance is always

- a) Negative
- b) Zero
- c) Positive

d) Vector

Answer: c) Positive

66. Displacement can be

a) Zero

b) Negative

c) Positive

d) All of these

Answer: d) All of these

67. Speed cannot be

a) Zero

b) Negative

c) Positive

d) Constant

Answer: b) Negative

68. Velocity can be zero when

a) Object is moving

b) Object returns to start

c) Distance is zero

d) Time is zero

Answer: b) Object returns to start

69. Acceleration is zero when

a) Velocity increases

b) Velocity decreases

c) Velocity is constant

d) Direction changes

Answer: c) Velocity is constant

70.Speed–time graph area gives

- a) Velocity
- b) Acceleration
- c) Distance travelled
- d) Force

Answer: c) Distance travelled

71.Distance–time graph slope gives

- a) Speed
- b) Force
- c) Mass
- d) Weight

Answer: a) Speed

72.Steeper distance–time graph means

- a) Lower speed
- b) Higher speed
- c) No speed
- d) Rest

Answer: b) Higher speed

73.A body at rest has

- a) Velocity
- b) Acceleration
- c) Zero speed
- d) Force

Answer: c) Zero speed

74.Motion always needs

- a) Force at all times

b) Surroundings for reference

c) Gravity

d) Energy only

Answer: b) Surroundings for reference

75. Newton is the unit of

a) Speed

b) Velocity

c) Force

d) Acceleration

Answer: c) Force

76. m/s is the unit of

a) Acceleration

b) Force

c) Speed

d) Mass

Answer: c) Speed

77. m/s^2 is the unit of

a) Velocity

b) Force

c) Acceleration

d) Speed

Answer: c) Acceleration

78. Vehicle moving at constant speed in straight line has

a) Acceleration

b) Uniform velocity

c) Non-uniform velocity

d) Deceleration

Answer: b) Uniform velocity

79. Bike moving in traffic shows

a) Uniform speed

b) Uniform velocity

c) Non-uniform velocity

d) Rest

Answer: c) Non-uniform velocity

80. Force can change

a) Motion of object

b) Shape of object

c) Direction of object

d) All of these

Answer: d) All of these